

Confidence Intervals for Population Proportions

Statistics Worksheet · Grade 11–12 / AP Statistics

Name: _____

Date: _____

Learning Objectives

- Verify the three conditions (randomness, normality via $np > 10$ and $nq > 10$, independence) before constructing a confidence interval for a population proportion
- Construct and interpret a confidence interval for a population proportion using the formula $\hat{p} \pm z^* \sqrt{\hat{p}\hat{q}/n}$
- Determine the minimum sample size required for a given margin of error and confidence level

Problems

1. A survey of 400 randomly selected voters found that 160 plan to vote in the upcoming election. Check whether the normality condition (rule of thumb) is satisfied for constructing a confidence interval for the population proportion.

$$n = 400, \quad \hat{p} = \frac{160}{400} = 0.40, \quad \hat{q} = 0.60$$

2. In a random sample of 250 high school students, 75 reported using social media for more than 5 hours per day. Find the sample proportion $\hat{p}$ and its complement $\hat{q}$.

$$\hat{p} = \frac{x}{n}, \quad \hat{q} = 1 - \hat{p}$$

3. Find the critical value z^* for a 90% confidence level using the formula for the critical value.

$$\alpha = 1 - C, \quad \frac{\alpha}{2} = \frac{1 - 0.90}{2}$$

4. A random sample of 500 adults found that 320 regularly exercise at least three times per week. Construct a 95% confidence interval for the true proportion of adults who exercise regularly. Use $z^* = 1.96$.

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

5. In a study on smartphone usage, 1,248 out of 4,800 randomly selected teenagers reported experiencing eye strain daily. Check all three conditions for constructing a confidence interval for the population proportion.



proportion, then identify the value of p-hat.

$$\hat{p} = \frac{1248}{4800}, \quad N \geq 10n$$

6. A 99% confidence interval for the proportion of college students who own a laptop is reported as (0.74, 0.86). Identify the sample proportion p-hat and the margin of error from this interval.

$$\hat{p} = \frac{\text{upper} + \text{lower}}{2}, \quad E = \frac{\text{upper} - \text{lower}}{2}$$

7. A health organization wants to estimate the proportion of adults who are physically inactive. They want their estimate to be within 4% at a 95% confidence level. They have no prior estimate of the proportion. Find the minimum sample size needed. Use $z^* = 1.96$ and p-hat = 0.50.

$$n \geq (z^*)^2 \cdot \frac{\hat{p}\hat{q}}{E^2}$$

8. A university poll of 2,200 randomly chosen undergraduates found that 594 had taken an online-only course in the past year. Construct a 99% confidence interval for the true proportion of undergraduates who have taken an online-only course. Use $z^* = 2.576$.

$$\hat{p} \pm 2.576 \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

9. A market research firm previously found that 35% of consumers preferred a particular brand. They want to update this estimate to within 2% at a 90% confidence level. Use $z^* = 1.645$ and p-hat = 0.35. What is the minimum sample size needed? How does this compare to the sample size required if no prior estimate were available (p-hat = 0.50)?

$$n \geq \frac{(z^*)^2 \hat{p}\hat{q}}{E^2}$$

10. A national polling agency surveyed 8,500 randomly selected registered voters and found that 3,740 supported a proposed education reform bill. (a) Verify all three conditions for a confidence interval for the population proportion. (b) Construct a 95% confidence interval using $z^* = 1.96$. (c) Write a conclusion interpreting the interval in context. (d) If the agency wants to conduct a follow-up survey accurate to within 1.5% at the 99% confidence level using the proportion found here, what is the minimum sample size needed? Use $z^* = 2.576$.



$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}\hat{q}}{n}}, \quad n \geq \frac{(z^*)^2 \hat{p}\hat{q}}{E^2}$$



Confidence Intervals for Population Proportions — Answer Key

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Answer Key

1. Answer: $np = 160 > 10$ and $nq = 240 > 10$; normality condition is satisfied

- Calculate $p = 160/400 = 0.40$ and $q = 1 - 0.40 = 0.60$
- Check $np = 400 \times 0.40 = 160 > 10$ ✓
- Check $nq = 400 \times 0.60 = 240 > 10$ ✓
- Both conditions are satisfied, so the normality condition holds.

2. Answer: $p = 0.30$, $q = 0.70$

- $p = 75 / 250 = 0.30$
- $q = 1 - 0.30 = 0.70$

3. Answer: $z^* = 1.645$

- $\alpha = 1 - 0.90 = 0.10$
- $\alpha/2 = 0.05$
- Look up the z-value that leaves 0.05 in each tail: $z^* = 1.645$

4. Answer: (0.598, 0.682)

- $p = 320/500 = 0.64$, $q = 0.36$
- Standard error = $\sqrt{(0.64 \times 0.36 / 500)} = \sqrt{(0.0004608)} \approx 0.02147$
- Margin of error = $1.96 \times 0.02147 \approx 0.0421$
- Lower bound = $0.64 - 0.0421 = 0.5979 \approx 0.598$
- Upper bound = $0.64 + 0.0421 = 0.6821 \approx 0.682$
- 95% CI: (0.598, 0.682)

5. Answer: $p = 0.26$; all three conditions satisfied (random ✓, $np = 1248 > 10$ ✓, $nq = 3552 > 10$ ✓, $N \geq 48,000$ ✓)

- Condition 1 – Randomness: stated as randomly selected ✓
- $p = 1248 / 4800 = 0.26$, $q = 0.74$
- Condition 2 – Normality: $np = 1248 > 10$ ✓ and $nq = 3552 > 10$ ✓
- Condition 3 – Independence: The total number of teenagers N must be $\geq 10 \times 4800 = 48,000$; this is very likely ✓
- $p = 0.26$

6. Answer: $p = 0.80$, $E = 0.06$

- $p = (0.74 + 0.86) / 2 = 1.60 / 2 = 0.80$
- $E = (0.86 - 0.74) / 2 = 0.12 / 2 = 0.06$

7. Answer: $n \geq 601$

- Use $p = q = 0.50$ (worst-case scenario when no prior estimate is available)



- $E = 0.04$, $z^* = 1.96$
- $n \geq (1.96)^2 \times (0.50 \times 0.50) / (0.04)^2$
- $n \geq 3.8416 \times 0.25 / 0.0016$
- $n \geq 0.9604 / 0.0016 = 600.25$
- Round up: $n \geq 601$

8. Answer: (0.244, 0.296)

- $p = 594 / 2200 = 0.27$, $q = 0.73$
- Standard error = $\sqrt{(0.27 \times 0.73 / 2200)} = \sqrt{(0.00008959)} \approx 0.009465$
- Margin of error = $2.576 \times 0.009465 \approx 0.02438$
- Lower bound = $0.27 - 0.02438 \approx 0.2456 \approx 0.246$
- Upper bound = $0.27 + 0.02438 \approx 0.2944 \approx 0.294$
- 99% CI: approximately (0.244, 0.296)

9. Answer: $n \geq 1,537$ using $p = 0.35$; $n \geq 1,692$ using $p = 0.50$; prior estimate reduces required sample size

- Using $p = 0.35$, $q = 0.65$, $z^* = 1.645$, $E = 0.02$:
- $n \geq (1.645)^2 \times (0.35 \times 0.65) / (0.02)^2$
- $n \geq 2.706025 \times 0.2275 / 0.0004 = 0.615620 / 0.0004 = 1539.05 \rightarrow n \geq 1,540$
- Using $p = 0.50$, $q = 0.50$:
- $n \geq 2.706025 \times 0.25 / 0.0004 = 0.676506 / 0.0004 = 1691.27 \rightarrow n \geq 1,692$
- Using the prior estimate of $p = 0.35$ requires a smaller sample (1,540 vs 1,692).

10. Answer: (a) All 3 conditions satisfied; (b) 95% CI: (0.430, 0.451); (c) We are 95% confident the true proportion of registered voters supporting the bill is between 43.0% and 45.1%; (d) $n \geq 7,212$

- (a) Condition 1: Randomly selected ✓; $p = 3740/8500 = 0.44$, $q = 0.56$
- Condition 2: $np = 3740 > 10$ ✓ and $nq = 4760 > 10$ ✓
- Condition 3: Total registered voters $N \gg 10 \times 8500 = 85,000$ ✓
- (b) $SE = \sqrt{(0.44 \times 0.56 / 8500)} = \sqrt{(0.00002898)} \approx 0.005384$
- $E = 1.96 \times 0.005384 \approx 0.01055$
- Lower = $0.44 - 0.01055 \approx 0.4295$; Upper = $0.44 + 0.01055 \approx 0.4505$
- 95% CI: (0.430, 0.451)
- (c) We are 95% confident the true proportion of registered voters who support the education reform bill is between 43.0% and 45.1%.
- (d) $n \geq (2.576)^2 \times (0.44 \times 0.56) / (0.015)^2$
- $n \geq 6.635776 \times 0.2464 / 0.000225 = 1.634833 / 0.000225 \approx 7,266 \rightarrow n \geq 7,266$

